

SE4010 – Cloud Computing Lab

Kafka Microservices on Azure Container Apps

Task 1 – Clone Repository

```
git clone https://github.com/seshangamage/Microservices-Architectural-Patterns.git
cd Microservices-Architectural-Patterns\Kaffka-Example
```

Task 2 – Azure Setup

```
az extension add --name containerapp --upgrade
az provider register --namespace Microsoft.App
az provider register --namespace Microsoft.OperationalInsights
az login
```

Task 3 – Variables

```
$RG="microservices-rg"
$LOC="westus2"
$ENV="micro-env"
$ACR="sliitmicroregistry"
$TAG="v1"
```

Task 4 – Create Resources

```
az group create `
--name $RG `
--location $LOC

az acr create `
--name $ACR `
--resource-group $RG `
--sku Basic `
--location $LOC

az acr update `
--name $ACR `
--admin-enabled true

az acr login `
--name $ACR

az containerapp env create `
--name $ENV `
--resource-group $RG `
--location $LOC
```

Task 5 – Build Services

```
cd gateway
mvn clean package -DskipTests
cd ..

cd order-service
mvn clean package -DskipTests
cd ..

cd inventory-service
mvn clean package -DskipTests
cd ..

cd notification-service
mvn clean package -DskipTests
cd ..
```

Task 6 – Build Docker Images

```
docker build -t gateway:v1 .\gateway
docker build -t order:v1 .\order-service
docker build -t inventory:v1 .\inventory-service
docker build -t notification:v1 .\notification-service
```

Task 7 – Tag + Push Images

```
docker tag gateway:v1 $ACR.azurecr.io/gateway:v1
docker tag order:v1 $ACR.azurecr.io/order:v1
docker tag inventory:v1 $ACR.azurecr.io/inventory:v1
docker tag notification:v1 $ACR.azurecr.io/notification:v1

docker push $ACR.azurecr.io/gateway:v1
docker push $ACR.azurecr.io/order:v1
docker push $ACR.azurecr.io/inventory:v1
docker push $ACR.azurecr.io/notification:v1
```

Task 8 – Deploy Kafka

```
az containerapp create `
--name kafka `
--resource-group $RG `
--environment $ENV `
--image confluentinc/cp-kafka:7.6.0 `
--ingress internal `
--target-port 9092
```

Task 9 – Deploy Backend Services

```
az containerapp create `
--name order-service `
```

```
--resource-group $RG `
--environment $ENV `
--image $ACR.azurecr.io/order:v1

az containerapp create `
--name inventory-service `
--resource-group $RG `
--environment $ENV `
--image $ACR.azurecr.io/inventory:v1

az containerapp create `
--name notification-service `
--resource-group $RG `
--environment $ENV `
--image $ACR.azurecr.io/notification:v1
```

Task 10 – Deploy Gateway

```
az containerapp create `
--name gateway `
--resource-group $RG `
--environment $ENV `
--image $ACR.azurecr.io/gateway:v1 `
--ingress external `
--target-port 8080
```

Task 11 – Get Gateway URL

```
$GATEWAY_FQDN = az containerapp show `
--name gateway `
--resource-group $RG `
--query properties.configuration.ingress.fqdn `
-o tsv

$GATEWAY_URL = "https://$GATEWAY_FQDN"
$GATEWAY_URL
```

Task 12 – Verify Deployment

```
az containerapp list `
--resource-group $RG `
--output table
```

Task 13 – Test API

```
Invoke-RestMethod `
-Uri "$GATEWAY_URL/orders" `
-Method Post `
-ContentType "application/json" `
-Body '{"orderId":"ORD-1001","sku":"SKU-ABC","quantity":2}'
```